

Tutorial 5: Clustering

Clustering of state and action spaces

November 25, 2024

- The tutorial aims to learn the basic concepts of unsupervised learning, another of the big fields of machine learning.
- Additionally, the tutorial will be used for the vector quantization technique of the last tutorial.
- The tutorial can be done in Linux, Windows or Mac.
- It is important to solve the exercises in order.

1 Introduction

In this tutorial, we will explore the fundamental concepts of unsupervised learning by working with clustering and dimensionality reduction techniques. Unsupervised learning enables us to uncover patterns in data without predefined labels. We will use two common approaches: clustering and principal component analysis (PCA). The exercises will be conducted using the scikit-learn library. The environment will be the same as in the previous tutorial, and the same data will be used.

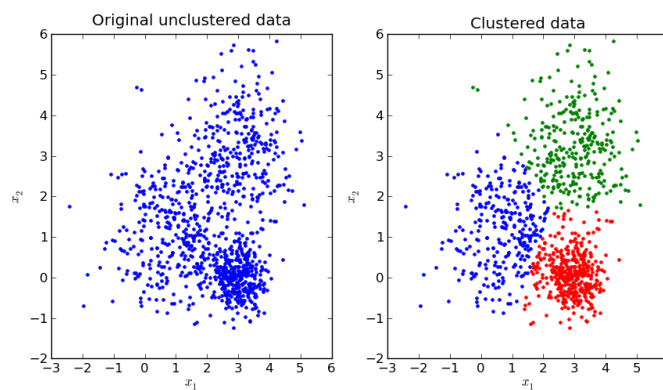


Figure 1: Clustering technique

2 Instructions

You should answer each question in the given order, indicating the exercise number. **The document should not contain screenshots with code.** The data for this tutorial will be the joined data from the previous tutorial. You can find it on Aula Global as *states_data.csv* and *actions_data.csv*.

3 Exercises

Exercise 1: K-Means Clustering

Clustering is a fundamental technique in unsupervised learning that groups data points into clusters based on similarity. In this exercise, we explore K-Means, a popular clustering algorithm called *KMeans*¹ in the scikit-learn library. In this case, the parameters to modify will simply be the number of clusters (*n_clusters*) and the initialization method (*init*). Additionally, we will add the normalization method to use between *MinMaxScaler*² and *StandardScaler*³. Since the actual label is not present, determining the best number of clusters is the key factor in this method. For that, we use increasing sequences such as 2, 4, 8, 16,...

1. Select a new set of parameters considering the mentioned attributes and perform a training process for the state space. Determine the inertia and silhouette score⁴ for each number of clusters. Determine the best number of clusters using the *elbow method* or the *silhouette method*.
2. Repeat the process for the action space. Do the best number of clusters differ from the state space to the action space? Why, or why not?
3. Visualize the clusters for the state space and the action space with their respective number of clusters and try to determine why are those clusters generated.

Exercise 2: Hierarchical Clustering

Unlike K-Means, hierarchical clustering builds a tree-like structure of clusters without needing a predefined number of groups. This exercise uses different linkage methods to analyze how hierarchical clustering performs on our data. The goal is to evaluate and visualize clustering results while understanding how the linkage criteria impact the final groupings. The hierarchical clustering method is called *AgglomerativeClustering*⁵ and the parameters to tune are the *linkage* method and the *distance.threshold*. For the parameter *n_clusters* we will set it to None. Finally, determine the normalization method as in the previous exercise.

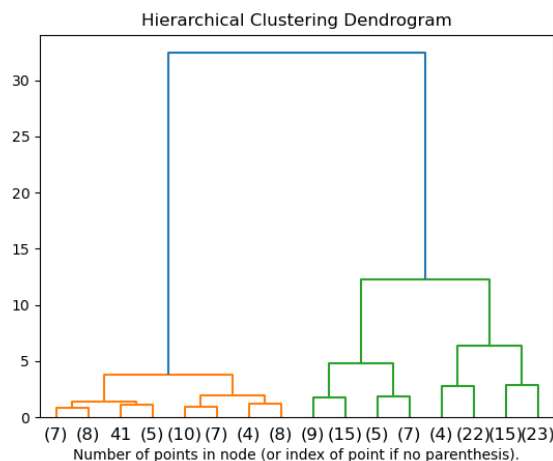


Figure 2: Hierarchical clustering

1. Select a set of parameters and perform a training process on the state space data. Determine the best number of clusters using the *Silhouette method*.
2. Repeat the process for the action space
3. Visualize the dendrograms for the action space with the best number of clusters. Interpret the results and determine why are those clusters selected.

¹<https://scikit-learn.org/1.5/modules/generated/sklearn.cluster.KMeans.html>

²<https://scikit-learn.org/1.5/modules/generated/sklearn.preprocessing.MinMaxScaler.html>

³<https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html>

⁴https://scikit-learn.org/stable/modules/generated/sklearn.metrics.silhouette_score.html

⁵<https://scikit-learn.org/stable/modules/generated/sklearn.cluster.AgglomerativeClustering.html>

Exercise 3: Principal Component Analysis (PCA)

When working with high-dimensional data, reducing the number of dimensions can simplify analysis and improve algorithm performance. This exercise focuses on PCA, a dimensionality reduction technique, to extract meaningful components while preserving as much variance as possible. The method is called *PCA*⁶ in the scikit-learn library. The only parameter is the number of components to use in the reduction. Apply a *StandardScaler* to the data.

1. Perform the PCA method on the state space using 3 as the number of components on the state space data. Plot the explained variance ratio for each principal component. How many components are necessary to capture at least 95% of the variance?
2. Perform again the PCA method with 2 as the number of components for the same data. Transform the data to be able to visualize it in 2D. Compare the visualization of the data in 2D and 3D. Do the spatial structures remain after the reduction?
3. Discuss how PCA could benefit clustering methods

4 Files to Submit

All the lab assignments **must** be done in groups of 2 people. You must submit a .zip file containing the required material through Aula Global before the following deadline: **Sunday, 8th December 2024 at 12:00**. The name of the zip file must contain the last 6 digits of both student's NIA, i.e., `tutorial5-123456-234567.zip`

The zip file must contain the following files:

1. A **PDF** document with (8 page limit without cover page):
 - Cover page with the names and NIAs of both students.
 - Answers to all the questions appearing in the Exercises section.
 - Description of the parameters used in each case are the selected values.
 - Explanation of the decisions made to select the best models.
 - Conclusions.
2. Include in a folder called "material" the following files:
 - The files used to implement all functionalities of the tutorial.

Please, **be very careful and respect the submission rules**.

⁶<https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html>